

Difference between Synchronous and Asynchronous Sequential Circuit

Synchronous Seq. Circuit

- ① These circuits are easy to design.
- ② A clocked FF acts as memory element.
- ③ They are slower.
- ④ The status of memory element is affected only at the active edge of clock, if I/P is changed.
- ⑤ Latches FFs are used in synchronous circuits.

Asynchronous Seq. Circuit

- ① These circuits are difficult to design.
- ② An unlocked FF or time delay element is used as memory element.
- ③ They are faster as clock is not present.
- ④ The status of memory element will change any time as soon as input is changed.
- ⑤ Latches are used in asynchronous circuits.